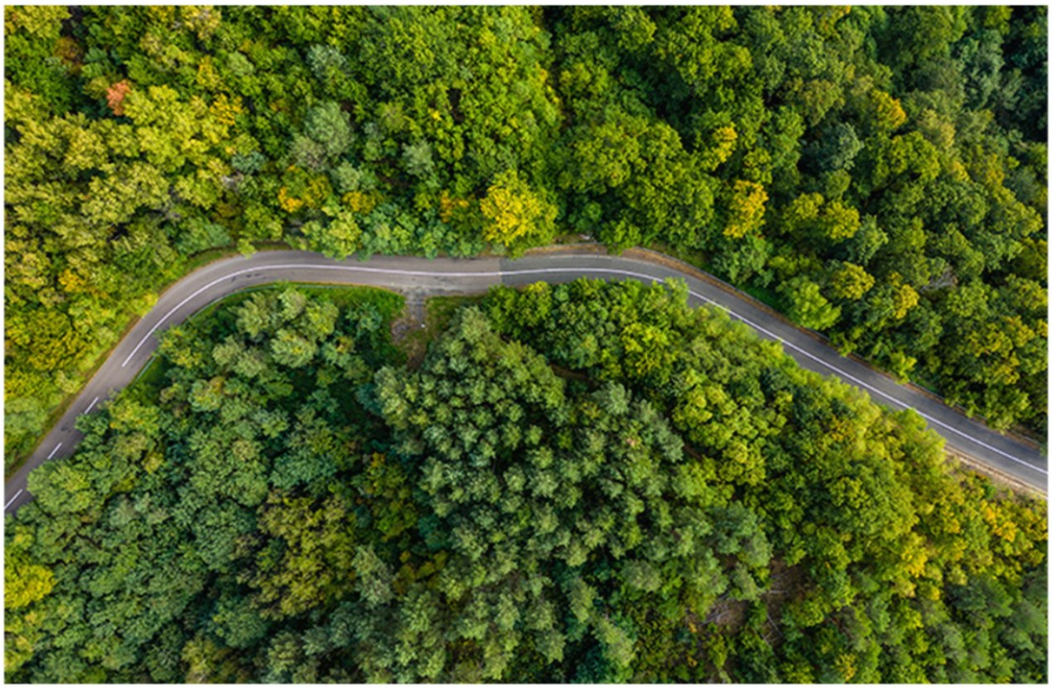

Forest fire detection with Deep Learning and satellite images

A NEW APPROACH FOR A NEAR REAL TIME DETECTION

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Exec Summary

Wildfires have been a major concern over last years because the change in their behavior, their speed and their destructive capacity.

Devastating Blazes

Disastrous blazes increased more than fourfold from 1980 to 2023 according to a paper in the journal Science. (Calum X. Cunningham1*, 2025). Based on that paper, disastrous wildfires occurred globally but were disproportionately concentrated in the Mediterranean and Temperate Conifer Forest biomes, and in populated regions that experience intense fire. Major economic losses caused by wildfire events increased by 4.4 fold from 1980 to 2023, amounting a 28.3 US$ billion and 0,03% of global GDP. Of the 200 most damaging events, 43% occurred in the last 10 years.

And the impact of the fires in on health should not be overlooked. Based on a paper on how "Wildfire smoke exposure" will increase mortality in USA over next decades (Minghao Qiu, 2025), that wildfire smoke would kill an estimated 70,000 Americans each year by 2050.

Disasters were heavily concentrated in the Mediterranean Forest/Woodland/Scrub Biome (Europe, southern South America, western USA, South Africa, and southern Australia) and the Temperate Conifer Forest Biome (mostly western North America), where disasters occurred 12.1 and 4.1 times more than expected based on the areas of those biomes, respectively.

On the other side, increasing expenditure on fire suppression has not prevented the rising occurrence of wildfire disasters. For example, US federal expenditure on fire suppression increased by 3.6 fold from 1985-2022 Therefore, it's of utmost importance the implementation of strategies that reduce transmission, NOT ONLY including retrofitting existing structures, using stringent fire-sensitive design and materials in new builds, establishing defendable space, and removing nearby fuel in the home ignition zone, BUT, as well, with a fast, early identification of fire spots or burned areas.

Wildfire monitoring.

In fact, wildfire season , in many regions, is year-round and continuous fire surveillance becomes critical to minimize potential impact in forests and communities. For a long time, watchtowers setup at hilltops have been the traditional means of wildfire monitoring, with an observation radius of up 20 km under ideal conditions (Rego, 2006). Then, automated image observation devices (e.g. video surveillance systems) mounted on watchtowers have largely replaced human observers, demonstrating superior performance during daylight hours (Günay, 2009). Even though, while automated thermal cameras enable basic 24-hour operation, there are still some limitations such visibility constraints from terrain or weather create permanent blind spots unreachable by static towers (Guangqing Zhai, 2025)

Thanks to the development of time series data, advancements in new sensor systems and image processing technology, and the increased availability of free images, Remote Sensing (RS) based on satellite and radar images has been widely applied in wildfire monitoring, especially suitable for vast remote areas ((Guangqing Zhai, 2025). It has been recognized that updated information on landscape fire activity is essential to aid fire management, and that information can only be provided by using satellite Earth Observation approaches (Wooster, 2021) This form of Earth observation is based on detecting the footprint or signature of the electromagnetic radiation emitted as biomass burns. Since the early 1980's active fire (AF) remote sensing conducted using Earth orbiting (LEO) satellites has been deployed in certain regions of the world to map the location and timing of landscape fire occurrence, and from the early 2000's global-scale information updated multiple times per day has been easily 100 available to all. Geostationary (GEO) satellites provide even higher frequency AF information, more than 100 times per day in some cases, and both LEO- and GEO-derived AF products now often include estimates of a fires characteristics, such as its fire radiative power output, in addition to the fires detection (Wooster, 2021). AF data provide information relevant to fire activity ongoing when the EO data were collected, and this can be delivered with very low latency times.

Satellite EO can be used to probe many fire characteristics, including burned area ((Giglio, 2018) and the concentration and composition of smoke plumes . Active fire (AF) remote sensing primarily focuses on identifying the location, timing and radiative strength (Fire Radiative Power; FRP) of fires that are actually consuming vegetation and/or organic soil at the time the observations were made.

Active fire (AF) detection and characterization is based on remote sensing of some of the approximately 20 MJ.kg-1 of energy released when vegetation and organic soil burns (Sullivan, 2008) Of the total energy released, only about 10 -20% is released as electromagnetic radiation. This radiative energy rate is far higher than from the same area of ambient land however, and its spectral distribution follows Planck's Radiation law and its derivative Wien's Displacement (Wooster, 2021). Based on this radiative analysis, there was a work to define Active Fire detection algorithms that could identify pixels that contain active fires. Those algorithms were defined to discriminate pixels with active fires from non-fire pixels, taken into consideration potential confusing effects such as sun glint, non-burning hot areas or regions with high local thermal contrast.

All in all, active fire detection has evolved over time, from reporting the timing and location of actively burning fires to include measures such as fire effective temperature, area and fire radiative power (FRP).

Using satellite images for fire detection

From Earth Observation (EO) satellites, we can get near real-time (NRT) data streams that can help on detection and monitoring of active fires. And there are expectations that future satellite mission, including higher spatial resolution Geostationary (GEO) systems and an increase in the number of Active Fire capable Low Earth Orbit (LEO) system, including those of small satellite constellations, will provide more opportunities to improve fire detection.

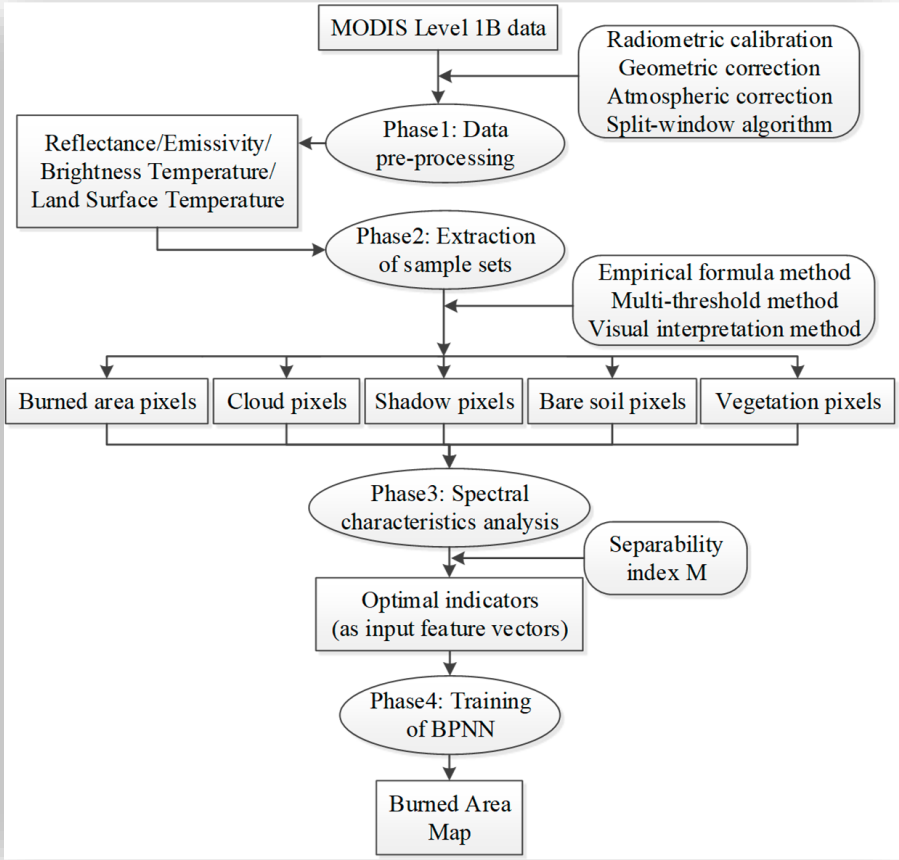
As of 2018, some authors started to map fires based on data from Sentinel-2 sensor (Fernando Rodriguez-Jimenez, 2023). The Sentinel-2 mission is part of the Copernicus Earth observation program, with twin satellites Sentinel 2A launched in 2015, and Sentinel 2B in 2017. Sentinel-2 carries multispectral imaging instruments on board. The spatial resolution for the output product chosen is 20 m, due to the spectral bands using the most usual indices for mapping burned areas and their severity. Fires are detected through the application of the Relativized Burn Ratio (RBR) (Sean Parks, 2014). This index is calculated from the Normalized Burn Ratio (NBR) using Band 8A and Band 12 in the pre-fire and post-fire images.

The new generation of geostationary satellites provides observations every 10 to 15 min at an improved spatial resolution (2–3 km) making it possible to detect short-lived fires not detectable by polar-orbiting satellites and to track in detail the evolution of the fire line and fire radiative power (Oliva, 2020). In addition, they have enhanced sensors that provide information on 12–16 spectral bands with improved radiometry of 10–14 bits. They introduce a substantial improvement in spatial, temporal, spectral, and radiometric resolution over their predecessors, which correspondingly relates to an enhanced capability for fire detection.

While an advantage of geostationary sensors is their temporal resolution, the polar-orbiting sensors have finer spatial resolution, which ensures higher accuracy at locating and mapping thermal anomalies. AF detections are produced using different polar-orbiting sensors, such as MODIS , the Visible Infrared Imaging Radiometer Suite (VIIRS) , FengYun-3C VIRR, Landsat-8 OLI , TET-1, and FireBird.

Currently, the best compromise between spatial and temporal resolution is provided by the operational active fire product derived from the 375 m VIIRS bands on board the National Polar-Orbiting Partnership (NPP) satellite since 2013 and on board the NOAA-20 since 2017. Its improved resolution, frequent acquisitions, and higher sensitivity to burning pixels allow the direct estimation of the burned area by aggregation of consecutive fire detections and the estimation of fire-driven deforestation .With two satellites in operation, the 375 m AF product time lapse between acquisitions is reduced to a few hours ensuring a higher frequency of observations providing fire behavior to fire managers and atmospheric modelers with higher resolution active fire data.

All in all, satellite earth observations are used more and more to monitor, estimate fire impacts by assessing the Burned Area (BA), their spatial variation as shape, size or orientation. An new algorithms have been developed for measuring Burned Area, incorporating machine learning approaches. Let's show an example using Moderate Resolution Imaging Spectroradiometer (MODIS) sensor image (Rui Ba, 2019), from the MODIS data of several wildfires in the American states of Nevada, Washington and California in 2016. They utilized a back-propagation neural network (BPNN) to learn the spectral differences of different types from the training sample sets and obtain the output burned area map.


MODIS Level 1B data
Radiometric calibration
Geometric correction
Atmospheric correction
Split-window algorithm
Phase1: Data pre-processing
Reflectance/Emissivity/ Brightness Temperature/ Land Surface Temperature
Phase2: Extraction of sample sets
Empirical formula method
Multi-threshold method
Visual interpretation method
Burned area pixels
Cloud pixels
Shadow pixels
Bare soil pixels
Vegetation pixels
Phase3: Spectral characteristics analysis
Separability index M
Optimal indicators (as input feature vectors)
Phase4: Training of BPNN
Burned Area Map

Our approach to Wildfire detection.

HYPOTHESIS:

Our aim with this project is to find the best fit in terms of datasets and deep learning architectures to the task of classifying and segmenting wildfires. Our expectation is to be closer as we can to a near real time detection of wildfires. Based on the previous research about which data can be used for fire detection, our hypothesis is that we can leverage satellite images from specific satellite sensors that are able to provide enough image resolution.

Imagine you have satellite photos of forests that caught fire. Your goal is to teach a computer to look at a satellite photo and say "fire happened here, here, and here" - like coloring in the burned areas. Once trained, our model can look at a satellite image it's never seen and draw the fire boundaries automatically.

WHERE GATHER DATA.

We have been looking at different sources of satellite data, and we have selected data from new sensors as Sentinel-2, because of its availability for medium-high spatial resolution of optical satellite data. This helps a more detailed assessment of small fires than other Burned Area products. Sentinel-2 is offered by the Copernicus Program, and offers a time series data of burned areas, with high revisit frequency, with improved spatial and spectral resolution of the MSI optical sensor (Filipponi, 2019)

Sentinel-2 is the most commonly used satellite for mapping the precise perimeter of a fire. They use SWIR (Short-Wave Infrared) Bands. While standard cameras only see smoke, Sentinel-2's Bands 11 and 12 can "see" the heat of the fire through the smoke. On a map, active fire fronts appear as glowing orange or red lines. (CEMS, Sentinel-2 Documentation, n.d.)

Then, they can run a burn Scar Analysis by comparing images from before and after a fire, scientists calculate the NBR (Normalized Burn Ratio). This identifies which areas were hit hardest and helps plan for forest restoration.

All in all, Sentinel-2 provides high-resolution images (10 to 20 meters), which is sharp enough to see individual roads, firebreaks, and structures

The satellite imagery comes from Sentinel-2 L2A, which spans across 12 distinct bands of the light spectrum, each with a resolution of 10 meters. The Sentinel Level-2A data undergoes an atmospheric correction process to adjust for the reflectance values influenced by the moisture in the atmosphere. These images are downloaded using SentinelHub APIs.

Then, we need to refer as well to Sentinel-3 (The "Thermometer in the Sky" (Global Monitoring) While Sentinel-2 takes a "snapshot" every few days, Sentinel-3 provides a broader daily view of the entire planet. It helps with Active Fire Detection. It uses a sensor called SLSTR that acts like a thermometer. It can detect "hotspots" and measure FRP (Fire Radiative Power)—a metric that tells firefighters how intense the fire is and how fast it consumes fuel. And it provides a World Fire Atlas.

Sentinel-3 data is used to maintain a global map of all active fires occurring at night, helping agencies track the spread of wildfires across entire continents.

DATASET SOURCES AND SELECTION

The Copernicus Emergency Management Service (CEMS) is a component of the European Union's Copernicus Programme. It provides rapid geospatial information during emergencies, and damage assessment for events such as floods, wildfires, and earthquakes. In particular, Copernicus Rapid Mapping provides on-demand mapping services in cases of various natural disasters, offering detailed and up-to-date geospatial information that assists in disaster management and risk assessment (CEMS, n.d.)

The Copernicus Emergency Management Service (CEMS) Wildfire dataset spans from June 2017 to April 2023. The dataset includes Sentinel-2 images related to wildfires, along with their respective severity and delineation masks. Additionally, the dataset is enhanced with cloud and landcover masks, providing more valuable information for future training of a semantic segmentation model. The dataset comprises over 500+ high-quality images, suitable for subsequent semantic segmentation model training. The dataset is available on Huggingface. (Hugginface, n.d.)

Dataset overview

Property	Value
Fire Events	~275 unique EMSR activations
Total images	500+ georeferenced tiles
Time Period	June 2017 - April 2023
Satellite	Sentinel-2 L2A (atmospherically corrected)
Resolution	~10 meters per pixel
Coverage	Europe (Portugal, Spain, Italy, Greece, etc.)
Format	GeoTIFF (.tif) + PNG previews

Here are the 12 bands from Sentinel-2 level-2A imagery

Band Index	Band Name	Wavelength	Resolution	Description
0	B01	443 (Coastal)	60m	Aerosol detection
1	B02	490 (Blue)	10m	Water, coastal
2	B03	560 (Green)	10m	Vegetation vigor
3	B04	665 (Red)	10m	Chlorophyll absorption

4	B05	705 (Red Edge 1)	20m	Vegetation stress
5	B06	740 (Red Edge 2)	20m	Vegetation stress
6	B07	783 (Red Edge 3)	20m	Leaf area index
7	B08	842 (NIR)	10m	Vegetation biomass
8	B09	865 (NIR narrow)	20m	Vegetation/water
9	B10	945 (Water Vapor)	60m	Atmospheric correction
10	B11	1610 (SWIR1)	20m	**Fire/burn detection**
11	B12	2190 (SWIR2)	20m	Fire/burn detection

From this table, we can define as Key bands for fire detection B11 and B12 (SWIR) since are the most sensitive to fire and burn scars.

Data format:

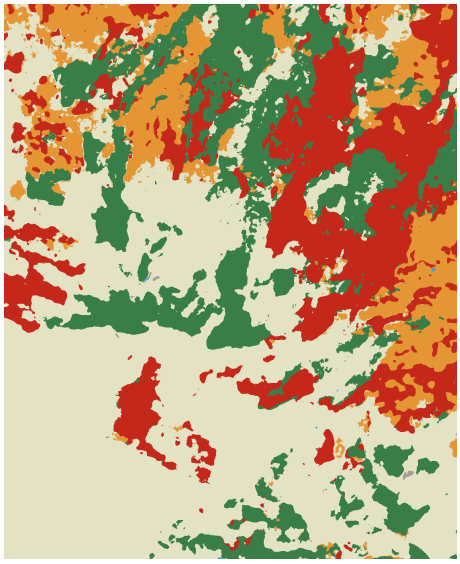
- Type: Float32
- Value range: 0.0 - 1.0 (surface reflectance)
- CRS: EPSG:4326 (WGS84)

Directory structure

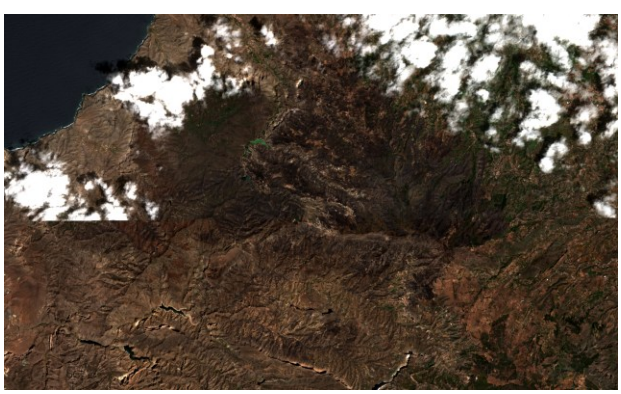
```
CEMS-Wildfire-Dataset/
├── csv_files/                  # Metadata files
│   ├── satelliteData.csv         # Main metadata (coordinates, dates, stats)
│   ├── datasetPreConfigured.csv    # Activation list with date ranges
│   └── cloudCoverage.csv         # Cloud/burn statistics per image
│
├── wildfires-cems/data/          # Image data (from HuggingFace)
│   ├── train/train/              # Training split
│   │   └── EMSR{XXX}/AOI{YY}/EMSR{XXX}_AOI{YY}_{ZZ}/
│   │       ├── *_S2L2A.tif        # Satellite image (12 bands)
│   │       ├── *_DEL.tif          # Delineation mask (binary)
│   │       ├── *_GRA.tif          # Grading mask (severity)
│   │       ├── *_CM.tif           # Cloud mask
│   │       ├── *_ESA_LC.tif       # Land cover (ESA WorldCover)
│   │       ├── *_S2L2A.json       # Acquisition metadata
│   │       └── *.png              # Preview images
│   ├── val/val/                  # Validation split
│   ├── test/test/                # Test split
│   └── cat/                      # Catalonia data (regional subset)
│
└── fire-pipeline/                # Processing tools
```

Data Structure and organization.

Here is sample of the dataset that gives an overview of the data structure and accompanying data. (Merlo, n.d.)



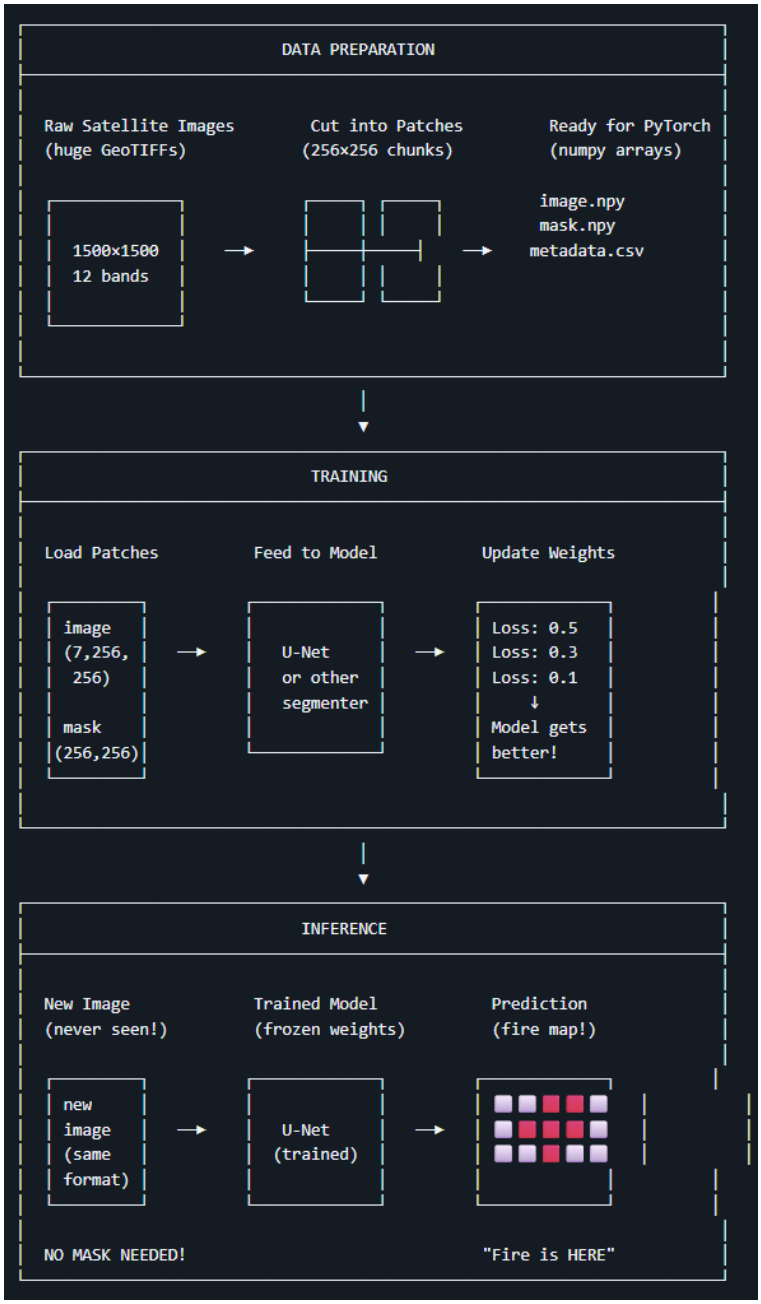

```
"headers": {
    "accept": "image/tiff",
    "content-type": "application/json"
},
"payload": {
    "evalscript": "//VERSION=3\nfunction setup() {\n    return {\n        input: [{\n            bands: [\"B01\", \"B02\", \"B03\", \"B04\", \"B05\", \"B06\", \"B07\", \"B08\", \"B8A\",
    "input": {
        "bounds": {
            "bbox": [
                -15.763778000999956,
                27.965664345000085,
                -15.649711279999963,
                28.090828702000058
            ],
            "properties": {
                "crs": "http://www.opengis.net/def/crs/EPSG/0/4326"
            }
        },
        "data": [
            {
                "dataFilter": {
                    "maxCloudCoverage": 10,
                    "mosaickingOrder": "leastCC",
                    "timeRange": {
                        "from": "2019-08-17T00:00:00Z",
                        "to": "2019-10-06T00:00:00Z"
                    }
                },
                "type": "sentinel-2-l2a"
            }
        ]
    },
    "output": {
        "height": 3380,
        "responses": [
            {
                "format": {
                    "type": "image/tiff"
                },
                "identifier": "default"
            }
        ],
        "width": 1129
    }
},
"timestamp": "2023-02-21T10:16:35.396520",
"url": "https://services.sentinel-hub.com/api/v1/process"
```

Experimental setup: Our approach to Wildfire detection with Deep Learning Architecture

[Experiment setup: here you explain what the experiment consists on (architecture, data...)]

THE COMPLETE PIPELINE



2 **Data loading and normalization**

3 **Sen2Fire reader** (fire-pipeline/sen2fire_dataset.py):

a. Load .npz: image (12, 512, 512), label (512, 512).

b. Normalize image: img = np.clip(img.astype(np.float32) / 10000., 0, 1).

c. Map to **7 channels** (or 8 with NDVI): indices [1,2,3,7,8,10,11]; optionally add NDVI from Red/NIR.

d. Output: **(8, 256, 256)** by center-cropping 256 from 512 (or 7 ch if include_ndvi=False).

4 **Patch size alignment (256 vs 512)**

a. **Center-crop to 256×256**: From each 512×512 patch, center 256×256 (image + label). Same architecture as CEMS.

5 **Cloud screening**

a. **Rule-based cloud score** (fire-pipeline/cloud_detection.py): cloud_score_sen2fire_12band(); in Sen2FireDataset, skip patches with cloud_score > max_cloud_score.

b. Optional: **s2cloudless** — get_cloud_fraction_s2cloudless() implemented; not wired into dataset (use when dependency added).

6 **Dataset and split mapping**

a. **Split mapping**: Train: scene1+scene2; Val: scene3; Test: scene4.

b. **Dataset class**: Sen2FireDataset(root_dir, split, max_cloud_score=..., include_ndvi=True) returns (image_tensor, mask_tensor).

7 **Training / fine-tuning**

a. **Fine-tune script** (train_sen2fire_finetune.py): Load CEMS checkpoint, freeze severity head, train binary on Sen2Fire; validate on scene3, optional test on scene4.

8 **Severity (GRA) handling**

a. **Dual-head**: Binary head (2) + severity head (5). Fine-tune on Sen2Fire with severity head **frozen**; only binary loss. App shows two toggleable layers (binary + severity).

DATA PREPARATION

What's in the data

Regular photos have 3 channels: Red, Green, Blue (RGB). But satellite images are special - they capture light our eyes can't see.

```
Normal Photo: 3 channels (RGB)
┌────┬─────┬────┐
│ Red │Green│Blue │
└────┴─────┴────┘

Satellite Image: 12 channels!
┌────┬─────┬────┬───┬────┬───┬─────┬─────┬───┬────┬────┬────┐
│Blue │Green│ Red │ ... │ NIR │ ... │SWIR1│SWIR2│ ... │
└────┴─────┴────┴───┴────┴───┴─────┴─────┴───┴────┴────┴────┘
  ▲                    ▲          ▲    ▲
  │                    │          └────┴── These "see" burned areas!
  │                    │                   (infrared light)
  └── Visible light    └── Near-infrared
      (what we see)        (plants reflect this)
```

Why so many channels? Different wavelengths reveal different things:

- SWIR (Short-Wave Infrared): Burned areas show up really clearly here - like a superpower for detecting fire damage
- NIR (Near-Infrared): Healthy plants reflect this strongly, dead/burned plants don't

Fire Masks (Labels)

For every satellite image, experts have drawn where the fire burned. This is your "answer key":

- **DEL (Delineation)** - Binary: burned or not burned
  - 0 = not burned (background)
  - 1 = burned (fire area)
- **GRA (Grading)** - Severity levels
  - 0 = no damage
  - 1 = minimal damage
  - 2 = moderate damage
  - 3 = high damage
  - 4 = destroyed

Why Do We Cut Images into Patches?

One of the challenges of satellite images is that they are huge. A single satellite image might be 1500 × 1500 pixels or bigger. On the other hand, neural networks:

- Need consistent input sizes (like 256 × 256)
- Would run out of memory with huge images
- Learn better from many small examples than few big ones

So, the solution is to slice the images up.

Why Overlap during training?

- A fire at the edge of patch 1 appears in the middle of patch 2
- The model sees each area from different "contexts"
- It's like data augmentation - more training examples!

In our case, we use 50% overlap (stride of 128 pixels) during training.

Which tasks can we tackle with all this data?

Based on the data we can use for our model, we can address 2 different type of tasks:

- Classification (Binary). Is the simpler task and it provides answer to the question "Does this patch contain any fire?
    - Yes (1)
    - No (0)
    - We could this by checking if any pixel in the mask > 0, then label = 1

- Segmentation. It addresses the question Which pixels are burned?. And the answer must provide a mask of the same size as the input. The Segmentation task is harder because you are making thousands of predictions (one per pixel) instead of just one for all the patch.

Understanding the Severity Mask (GRA)

When opening the TIF files from the CEMS dataset, that file contains integer values 0,1,2,3,4 and it looks that all the image is black. This has an explanation, since in a scale of 256 colors, 0-4 are near zero, that means near to black. So, the data is correct, it just needs some scaling.

In fact, the PNG file is a visualization of that TIF with a colormap applied to those 0-4 values:

Value	Meaning	PNG Color	RGB
0	No damage	Black	(0,0,0)
1	Negligible	Light Green	(181, 254, 142)
2	Moderate	Yellow	(254, 217, 142)
3	High	Orange	(254, 153, 41)
4	Destroyed	Dark Red	(204, 76, 2)

So, for training, use the TIF (with values 0-4). The PNG is just for humans to look at.

Not all images have severity (GRA)

Some fires only have DEL (binary burned/not burned), while some have both DEL and GRA (severity levels). This depends on what CEMS analysts provided. The satelliteData.csv has a GRA column (0 or 1) indicating availability.

Why Cloud masks matter

When the satellite takes a photo, it captures whatever is there - including clouds. Since The cloud pixels in contain cloud reflectance, NOT ground information. That data is useless for fire detection.

Cloud masks help since we can ignore those pixels marked as cloud reflectance. Then, Either we skip those pixels or we mask out cloudy pixels from the loss calculation. Therefore, during training we skip patches with too many clouds (> 50% cloudy is unreliable)

On the other side, during inference, we must flag predictions in cloudy areas as "uncertain" or, just don't make predictions for those pixels.

Data Splits

The dataset is pre-split into train/val/test:

SPLIT	LOCATION	PURPOSE
TRAIN/TRAIN/	Training data	Model training
VAL/VAL/	Validation data	Hyperparameter tuning
TEST/TEST/	Test data	Final evaluation
CAT/	Catalonia subset	Regional testing (optional)

To get statistics per split:

```
from pathlib import Path

data_root = Path("wildfires-cems/data")

for split in ["train", "val", "test"]:
    split_dir = data_root / split / split
    if split_dir.exists():
        emsr_folders = list(split_dir.glob("EMSR*"))
        print(f"{split}: {len(emsr_folders)} EMSR activations")
```

Wildfire masks

From Copernicus Rapid Mapping for each activation under the tag Wildfire are available different post-fire products:

- FEP (First Estimation): a first estimation of the burned area.
- DEL (Delineation): a delineation of the area affected by the wildfire.
- GRA (Grading): a detailed description about the severity of the burned area.

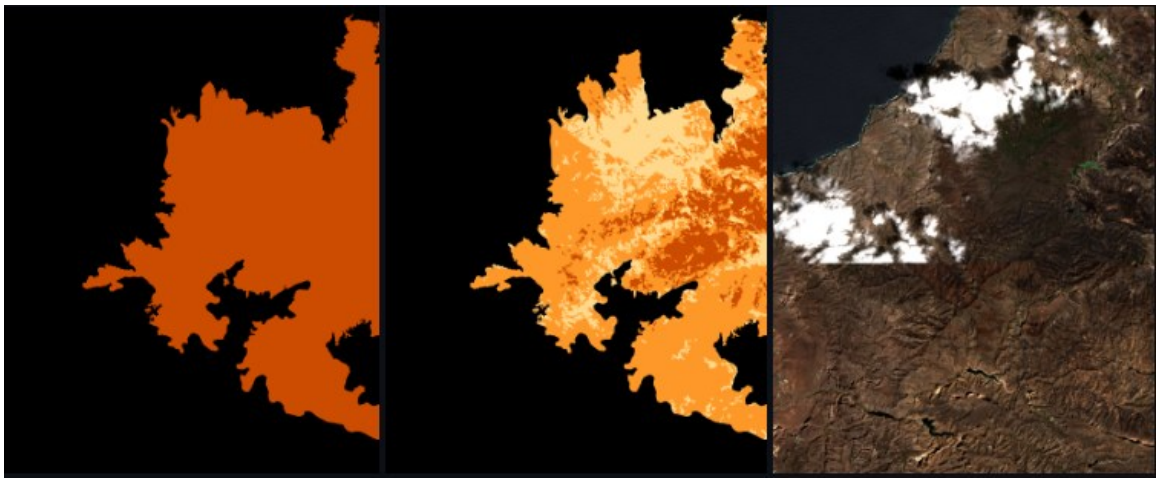
Each product includes metadata and associated JSON files which contain geographical details about the affected areas (you can see above an excerpt of a json file)

On Copernicus site are available several georeferenced data in GeoJSON format. For this dataset are considered only those files for each activation that are formatted with the following string:

- EMSRXXX_AOIYY_TYPE_PRODUCT_areaOfInterestA.json, where TYPE can be GRA, DEL or FEP, defines the AOI YY of that particular activation XXX where the event happened.
- EMSRXXX_AOIYY_TYPE_PRODUCT_observedEventA.json, where TYPE can be DEL or FEP, defines the multipolygons geometry for the wildfire delineation for an AOI YY of the activation XXX.
- EMSRXXX_AOIYY_GRA_PRODUCT_naturalLandUseA.json defines the various multipolygons geometry for the grading damage levels for an AOI YY of the activation XXX

Those are the different grading levels of damage used in the Copernicus products:

Damage Level	CopernicusEMS class	EMS-98 class	Color
0	No visible damage	No visible damage	
1	-	Negligible to slight damage	
2	Possibly damaged	Moderate damage	
3	Damaged	High damage	
4	Destroyed	Destruction	



Here in order are reported the DEL map, GRA map and the actual sentinel-2 Image for the activation EMSR382.

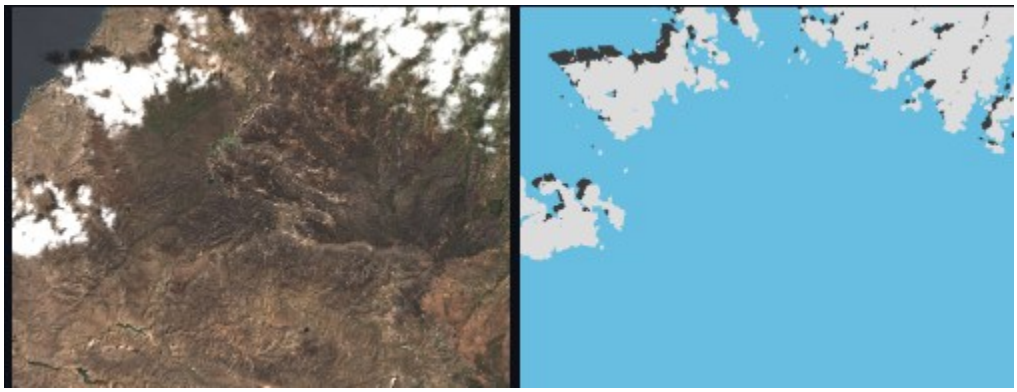
How manage cloud masks.

A special topic is on how deal with clouds (a weather condition) in the images from the dataset. Creating cloud masks before making inferences on Sentinel-2 images is important because clouds can obscure or distort the underlying land cover or land use information that is the focus of the analysis. This can lead to inaccurate or incomplete results. Sentinel-2 images are often used for remote sensing applications, such as monitoring vegetation health, mapping land cover and land use, and detecting changes over time. However, clouds can interfere with these applications by blocking or reflecting the light that is captured by the satellite, which can result in missing or distorted data. By default, all images are retrieved from sentinel-hub with the condition of no more than 10 percent of cloud coverage. However some images have a relevant cloud coverage.

This dataset makes available for each image a cloud masks: the areas that are affected by clouds can be identified and excluded from future analyses. This ensures that the inferences made from the Sentinel-2 data are based on accurate and reliable information. The masks were generated using the CloudSen12 model.

The output prediction of CloudSen12 has 4 different layers for cloud coverage:

Label	Class	Class definitions	Color
0	`Clear`	Areas where no clouds are found	
1	`Clouds`	Clouds and heavy clouds are present, terrain is obscured and not visible	
2	`Light clouds`	Areas affected by light clouds, where could cause some issue in the terrain visibility. In this class are included also fog and wildfire's smoke.	
3	`Shadow`	This areas are in the shadow of the clouds. The terrain is partially/fully visible but the color and some bands of sentinel2 could be changed from real value.	



he cloud masks resulted from CloudSen12 model for activation EMSR382

Smoothing-blend image patches

Let's start taking into consideration how data should be prepared for selected deep learning models. One challenge of using a U-Net architecture for image segmentation is to have smooth predictions, especially if the receptive field of the neural network is a small amount of pixels. In the context of the U-Net architecture for image segmentation, blending image patches can be used to generate smooth predictions by reducing the effect of discontinuities at patch boundaries. This approach involves dividing the input image into overlapping patches, running the U-Net architecture on each patch individually, and then blending the resulting predictions together to form a single output image.

By blending the predictions from multiple patches, the resulting output image is typically smoother and more continuous than if a single U-Net model was trained on the entire input image. This can help to reduce artifacts and improve the overall quality of the segmentation results. In this work the source code of cloudSen12 has been customized so that it could be smoothly predicted. (Chevalier, n.d.)

Landcover masks

In addition to wildfire delineation, severity and cloud masks, also the landcovers is provided for each image. In particular the models considered are:

- **ESRI 10m Annual Land Use Land Cover (2017-2021)**;
- **ESRI 2020 Global Land Use Land Cover**;
- **ESA WorldCover 10 m 2020**.

All this landcover are downloaded from Planetary Computer. All landcover models are based on sentinel2 10-meter resolution.

ESA WorldCover 2020

File: *_ESA_LC.tif

Value	Class
10	Tree cover
20	Shrubland
30	Grassland
40	Cropland
50	Built-up
60	Bare/sparse vegetation
70	Snow and ice
80	Permanent water bodies
90	Herbaceous wetland
95	Mangroves
100	Moss and lichen

Metadata Files

satelliteData.csv

Location: csv_files/satelliteData.csv **Rows**: ~560 (one per image tile)

The primary metadata file containing geographic, temporal, and statistical information.

Column	Type	Description
EMSR	string	Emergency activation ID (e.g., "EMSR207")
AOI	string	Area of Interest (e.g., "AOI01")
folder	string	Data quality category: "optimal" or "colomba"

folderPath	string	Relative path to image directory
activationDate	datetime	When CEMS was activated for this fire
interval_startDate	datetime	Start of Sentinel-2 search window
interval_endDate	datetime	End of Sentinel-2 search window
post_fire_acquisition	string	Actual satellite acquisition timestamp
GRA	bool (0/1)	Grading mask available
DEL	bool (0/1)	Delineation mask available
FEP	bool (0/1)	First Estimate Product available
left_Long	float	Bounding box: left longitude
bottom_Lat	float	Bounding box: bottom latitude
right_Long	float	Bounding box: right longitude
top_Lat	float	Bounding box: top latitude
centerBoxLong	float	Center longitude
centerBoxLat	float	Center latitude
resolution_x	float	Pixel size in X (degrees)
resolution_y	float	Pixel size in Y (degrees)
height	int	Image height in pixels
width	int	Image width in pixels
pixelBurned	int	Number of burned pixels in DEL mask
country	string	Country name (Portugal, Spain, Italy, etc.)
koppen_group	string	Köppen climate classification group

koppen_subgroup	string	Köppen climate subgroup

Example query - Find all fires in Spain:

```
import pandas as pd
df = pd.read_csv("csv_files/satelliteData.csv")
spain_fires = df[df["country"] == "Spain"]
```

datasetPreConfigured.csv

Location: csv_files/datasetPreConfigured.csv Rows: ~274 (one per EMSR+AOI combination)

High-level activation information for downloading/configuring the dataset.

Column	Type	Description
EMSR	string	Emergency activation ID
AOI	string	Area of Interest
folderType	string	"optimal", "subOptimal_cloudy", or "subOptimal_FEP"
folderPath	string	Base folder path
activationDate	datetime	When CEMS was activated
suggested_startDate	datetime	Recommended search window start
suggested_endDate	datetime	Recommended search window end

cloudCoverage.csv

Location: csv_files/cloudCoverage.csv **Rows**: ~998 (entries for both DEL and GRA masks)

Statistics about cloud cover and burn extent per image.

Column	Type	Description
EMSR_AOI	string	Combined identifier (e.g., "EMSR230_AOI01_01")

folderPath	string	Path to image directory
startDate	datetime	Search window start
endDate	datetime	Search window end
height	int	Image height
width	int	Image width
sizeImage	int	Total pixels (height × width)
burnedPixel	int	Pixels with burn mask > 0
cloudPixel	int	Pixels with cloud mask > 0
countOverlap	int	Pixels where burn and cloud overlap
percentageCloud	float	Fraction of image with clouds
percentageOverlap	float	Fraction of burned area obscured by clouds
Type	string	"DEL" or "GRA"

JSON Metadata

Contains Sentinel Hub API request details and acquisition metadata

```
{
  "payload": {
    "input": {
      "bounds": {
        "bbox": [lon_min, lat_min, lon_max, lat_max],
        "properties": {"crs": "EPSG:4326"}
      },
      "data": [{
        "dataFilter": {
          "maxCloudCoverage": 10,
          "timeRange": {"from": "...", "to": "..."}
        },
        "type": "sentinel-2-l2a"
      }]
    },
    "output": {"height": 580, "width": 589},
    "acquisition_date": ["2017/08/15_09:31:03"]
```

```
  },
  "timestamp": "2023-04-23T01:51:47.114008"
}
```

Key fields:

- bbox: Exact bounding box coordinates
- acquisition_date: When the satellite captured the image
- timeRange: Search window used to find the image

How Everything Connects

```
                    ┌──────────────────────────────────┐
                    │        datasetPreConfigured.csv  │
                    │   (activation dates, download config)  │
                    └──────────────────────────────────┘
                                   │
                                   │
                                   │ EMSR + AOI
                                   ▼
┌──────────────────────────────────────────────────────────────────┐
│                         satelliteData.csv                        │
│  (coordinates, country, climate, resolution, burned pixels, dates)  │
└──────────────────────────────────────────────────────────────────┘
                                   │
                                   │
                                   │ folderPath
                                   ▼
┌──────────────────────────────────────────────────────────────────┐
│                       Image Directory                            │
│  wildfires-cems/data/{split}/{split}/EMSR{XXX}/AOI{YY}/EMSR{XXX}_...  │
│                                                                  │
│   ├── *_S2L2A.tif    ◄── Input (satellite imagery)               │
│   ├── *_S2L2A.json   ◄── Acquisition metadata                    │
│   ├── *_DEL.tif      ◄── Target (binary mask)                    │
│   ├── *_GRA.tif      ◄── Target (severity mask)                  │
│   ├── *_CM.tif       ◄── Filter (cloud mask)                     │
│   └── *_ESA_LC.tif   ◄── Context (land cover)                    │
└──────────────────────────────────────────────────────────────────┘
                                   │
                                   │ EMSR_AOI identifier
                                   ▼
                    ┌──────────────────────────────────┐
                    │          cloudCoverage.csv       │
                    │   (cloud %, burn %, overlap stats)  │
                    └──────────────────────────────────┘
```

Joining data example:

```
import pandas as pd

# Load metadata
satellite_df = pd.read_csv("csv_files/satelliteData.csv")
cloud_df = pd.read_csv("csv_files/cloudCoverage.csv")

# Create matching identifier in satellite_df
satellite_df["EMSR_AOI"] = satellite_df.apply(
    lambda r: f"{r['EMSR']}_{r['AOI']}_01", axis=1  # Adjust tile number as
needed
)

# Join on identifier
merged = satellite_df.merge(
    cloud_df[cloud_df["Type"] == "DEL"],
    on="EMSR_AOI",
    how="left"
)
```

Common use cases with data

1. Binary File Segmentation

```
# Load image and mask
import rasterio

with rasterio.open("path/to/EMSR230_AOI01_01_S2L2A.tif") as src:
    image = src.read()  # Shape: (12, H, W)

with rasterio.open("path/to/EMSR230_AOI01_01_DEL.tif") as src:
    mask = src.read(1)  # Shape: (H, W), values 0 or 1
```

2. Filter by Country / Region

```
import pandas as pd

df = pd.read_csv("csv_files/satelliteData.csv")

# Get Portuguese fires only
portugal = df[df["country"] == "Portugal"]

# Get fires in a specific bounding box (e.g., Catalonia)
catalonia = df[
    (df["country"] == "Spain") &
    (df["centerBoxLong"] >= 0.15) & (df["centerBoxLong"] <= 3.35) &
    (df["centerBoxLat"] >= 40.5) & (df["centerBoxLat"] <= 42.9)
]
```

3. Filter by Data Range

```
df["activationDate"] = pd.to_datetime(df["activationDate"])

# Fires from 2022
fires_2022 = df[df["activationDate"].dt.year == 2022]

# Summer fires (June-September)
summer_fires = df[df["activationDate"].dt.month.isin([6, 7, 8, 9])]
```

4. Filter by Burn Size

```
# Large fires only (>100,000 burned pixels)
large_fires = df[df["pixelBurned"] > 100000]

# Calculate actual area (approximate)
# At 10m resolution: area = pixelBurned * 10 * 10 = pixelBurned * 100 m²
df["burned_area_km2"] = df["pixelBurned"] * 100 / 1e6
```

5. Exclude Cloud images

```
cloud_df = pd.read_csv("csv_files/cloudCoverage.csv")

# Keep only images with <10% cloud cover
clear_images = cloud_df[cloud_df["percentageCloud"] < 0.10]
```

Sen2Fire Dataset Integration Plan

Here we provide some outlines about how to add the Sen2Fire dataset (Australia, binary fire only, no cloud masks) to the existing CEMS-based pipeline, and how to handle clouds and training strategy.

Component	Location	Notes
Dual-head model	fire-pipeline/model.py	FireDualHeadModel: binary head (2 classes) + severity head (5). freeze_severity_head() for Sen2Fire fine-tune.
CEMS dual-head training	fire-pipeline/train.py	--dual-head (requires --num-classes 5 and GRA patches). Trains both heads; saves dual_head: true in config.
Sen2Fire dataset	fire-pipeline/sen2fire_dataset.py	Sen2FireDataset: loads .npz, 7/8 ch, center-crop 256. **Cloud filter**: s2cloudless by default when installed, else rule-based (max_cloud_score). use_s2cloudless=True (default); --

		no-s2cloudless in fine-tune script for rule-based only. Splits: train=scene1+2, val=scene3, test=scene4.
Cloud detection	fire-pipeline/cloud_detection.py	Rule-based cloud_score_*; get_cloud_fraction_s2cloudless() and get_cloud_fraction_sen2fire() (10-band subset for s2cloudless). Optional dep: s2cloudless in train / all extras.
Sen2Fire fine-tune	fire-pipeline/train_sen2fire_finetune.py	Loads CEMS checkpoint (dual or single-head), freezes severity head, trains binary on Sen2Fire. --checkpoint, --sen2fire-dir, --output-dir, --max-cloud-score, --no-s2cloudless, --no-ndvi.
Inference dual-head	fire-pipeline/inference.py	Loads FireDualHeadModel when config.dual_head; returns InferenceResult with binary_segmentation, severity_segmentation, binary_probabilities, severity_probabilities.
App: two layers	fire-pipeline/app.py	When result.dual_head, two checkboxes: "Show binary fire map", "Show severity map"; each toggles a separate overlay.
History (dual-head)	fire-pipeline/storage.py	Saves/loads binary_segmentation, severity_segmentation, binary_probabilities, severity_probabilities in result.npz; metadata.dual_head so history view shows both layers.
Tests	fire-pipeline/tests/	test_cloud_detection.py, test_sen2fire_dataset.py, test_model_dual_head.py; dual-head save/load in test_storage.py; dual-head InferenceResult and create_visualization_from_segmentation in test_inference.py.

Sen2Fire dataset summary

Aspect	**Detail**
Source	Zenodo 10881058, paper: Sen2Fire arXiv:2403.17884
Content	2,466 patches from 4 bushfire scenes, NSW Australia, 2019–2020 season

Patch size	512×512 (vs CEMS 256×256)
Bands	12 Sentinel-2 L2A + 1 Sentinel-5P aerosol index (13 channels)
Labels	Binary fire mask only (0/1), from MOD14A1 V6.1 — **no severity (GRA)**
Splits	**Train**: scene1 + scene2 (1,458 patches). **Val**: scene3 (504). **Test**: scene4 (504).
Format	.npz per patch: image (12, 512, 512) int16, aerosol (512, 512), label (512, 512) uint8
Values	Reflectance-style DN (e.g. ×10000); need to normalize to [0, 1] or similar
Cloud mask	**None** — we need to infer or filter cloudy patches

THE TRAINING PROCESS

- **Goal**: Train a **semantic segmentation** model on CEMS wildfire patches (7-channel input, pixel-wise labels) and optionally use it for **binary fire detection** (patch-level "has fire" derived from segmentation).
- **Entry point**: train.py in fire-pipeline/. It builds data loaders (from the data transformation pipeline), model, loss, optimizer, scheduler, and metrics; runs train/val loops; saves checkpoints and supports resume and W&B.
- **Outputs**: output_dir/config.json, output_dir/checkpoints/best_model.pt, final_model.pt, and periodic checkpoints.

One training script covers both DEL (binary) and GRA (5-class); we pick the best model by validation fire IoU and can resume or log to Weights&Biases.

For the training process we plan to use an input Tensor with the following shape:

- Multiple images at once (batch)
- 7 spectral bands (we will pick the useful ones from here)
- Width and Height of 256 x 256

On the other side, the shape of the Target mask will include:

- One label per pixel (0,1,2,3 or 4), representing the fire severity.
- Same Width and Height as input: 256 x 256

Model

Architecture

- Goal: Train a semantic segmentation model on CEMS wildfire patches (7-channel input, pixel-wise labels) and optionally use it for binary fire detection (patch-level "has fire" derived from segmentation).
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- Outputs: output_dir/config.json, output_dir/checkpoints/best_model.pt, final_model.pt, and periodic checkpoints.

FireSegmentationModel wrapper

A single model does pixel-wise segmentation and patch-level detection/confidence. e don't train a separate classifier; detection is derived from segmentation by 'any fire pixel in the patch

- Goal: Train a semantic segmentation model on CEMS wildfire patches (7-channel input, pixel-wise labels) and optionally use it for binary fire detection (patch-level "has fire" derived from segmentation).
- Entry point: train.py in fire-pipeline/. It builds data loaders (from the data transformation pipeline), model, loss, optimizer, scheduler, and metrics; runs train/val loops; saves checkpoints and supports resume and W&B.
- Outputs: output_dir/config.json, output_dir/checkpoints/best_model.pt, final_model.pt, and periodic checkpoints.

Loss

We combine CE and Dice and can add class weights or Focal to handle rare fire pixels

CombinedLoss

- **Formula**: total = 0.5 * CE + 0.5 * Dice.
- **CE**: Pixel-wise cross-entropy; optional **class weights** (inverse frequency); optional **Focal** variant (γ=2) instead of plain CE.
- **Dice**: 1 – mean(Dice per class); smooth term 1e-6. Helps with **imbalanced** fire pixels (region-based).
- **Returns**: (total_loss, dict with ce_loss, dice_loss) for logging.

Class weights and Focal

- **Class weights**: If use_class_weights=True, weights are computed from the **training** split (compute_class_weights(patches_dir/train, num_classes)) and passed to CE (or Focal). Fire class(es) get higher weight.
- **Focal**: --focal-loss switches CE to Focal (same weights, γ=2 by default). Down-weights easy pixels and focuses on hard ones.

Optimizer and Scheduler

- **Optimizer**: **AdamW** (lr=1e-4, weight_decay=1e-4).
- **Scheduler**: **ReduceLROnPlateau** on **validation fire_iou** (mode=max): factor=0.5, patience=5. So LR is reduced when val fire IoU stops improving.

What the model learns

The model learns patterns like:

- "When SWIR bands show high values + NIR shows low values → probably burned"
- "This texture pattern usually means fire damage"
- "Edges between these colors often indicate burn boundaries"

Sen2fire Fine-tune

1. Train dual-head on CEMS (GRA patches, NDVI default)

   uv run python train.py --patches-dir ./patches --output-dir ./output/dual --num-classes 5 --dual-head

2 Fine-tune on Sen2Fire (binary only, severity frozen; s2cloudless used by default)

   uv run python train_sen2fire_finetune.py \

   --checkpoint ./output/dual/checkpoints/best_model.pt \

   --sen2fire-dir ../data-sen2fire \

   --output-dir ./output/sen2fire_ft

Our training strategy: CEMS first, then SEN2Fire

Recommendation: train on CEMS → fine-tune on Sen2Fire (with optional joint training variant)

Approach	**Pros**	**Cons**
CEMS only	Already implemented	No Australia / geographic diversity
Sen2Fire only	Simple	Small (1,458 train), single region, no severity
Joint from scratch	One pipeline	Different formats/sizes, label mismatch (severity vs binary), risk of CEMS dominance
CEMS train → Sen2Fire fine-tune ✓	Reuse CEMS representation; add Australian data; binary head matches Sen2Fire	Need a clear fine-tune protocol
Joint with two loaders	Can balance datasets	More complex; need to align input channels and loss (e.g. binary-only on Sen2Fire)

Trainin Plan (1):

1. **Phase 1 – CEMS (current)**
   - Train segmentation model on CEMS (binary DEL or 5-class GRA, as now).

    - Output: pretrained backbone + binary (and optionally severity) head.

2. **Phase 2 – Fine-tune on Sen2Fire**
    - Use **binary fire only** (Sen2Fire has no severity).
    - Options:
        - **A)** Freeze encoder, train only decoder/head on Sen2Fire (fast, preserves CEMS features).
        - **B)** Full fine-tune with small LR (e.g. 1e-5) to adapt to Australian conditions.
    - Input: map Sen2Fire 12 (+ aerosol) to same **7 or 8 channels** as CEMS (same 7 bands; add NDVI for 8-channel to match current CEMS default).
    - Patch size: either crop 512→256 from center (or multiple 256×256 crops per 512 patch) so the model sees 256×256 as in CEMS.
3. **Evaluation**
    - CEMS val/test for European performance.
    - Sen2Fire val (scene3) during fine-tune; Sen2Fire test (scene4) for final Australian / out-of-region report.
4. **Optional later**
    - Joint training: two dataloaders (CEMS + Sen2Fire), binary loss on both; severity loss only on CEMS (ignore severity on Sen2Fire). Requires consistent 7-channel input and 256×256 patches from both.

Cloud handling for Sen2Fire (no cloud masks)

Because Sen2Fire has **no cloud mask**, we have three practical options.

Option A – Rule-based cloud screening (recommended first)

Use spectral heuristics to flag "likely cloudy" patches and **exclude them from training** (and optionally from val/test).

- **Idea**: Clouds are bright in visible (e.g. Blue, Green), often flat in NIR, and we have Water Vapour (B10) which is high over clouds.
- **Simple proxy**: e.g. mean reflectance in B02 (Blue) or (B02+B03)/2 above a threshold → "cloudy".
- **Alternative**: Compute a **simple cloud score** per patch, e.g.

    - cloud_score = mean(B02 + B03) / (1 + mean(B08))
    - or use B10 (water vapour) if available in the 12-band cube.

- **Action**: Exclude patches with cloud_score > threshold (tune threshold on a few manual checks or on CEMS by comparing to real CM).
- **Pros**: No extra model, no external API, fast.
- **Cons**: Heuristic; may drop some usable patches or keep some cloudy ones.

Option B – s2cloudless (or similar) cloud mask

- Use **s2cloudless** (Sentinel Hub) or another Sentinel-2 cloud detector to predict a cloud mask per patch.
- Then: either **drop patches** with cloud fraction > X%, or **mask out** cloudy pixels in the loss (same idea as CEMS).
- **Pros**: Better accuracy than a simple rule.
- **Cons**: Extra dependency; possibly different resolution; need to run it over all Sen2Fire patches once and save masks or exclude list.

Option C – Cloud classifier trained on CEMS

- Train a small **binary cloud classifier** on CEMS (input: 7-channel patch, label: from CEMS CM).
- Run this classifier on Sen2Fire patches; treat "cloud probability" as cloud fraction and exclude (or mask) high-cloud patches.
- **Pros**: Same sensor/bands as your pipeline; no external service.
- **Cons**: Need to implement and train; CEMS cloud distribution may differ from Australia.

Recommendation: Start with **Option A** (rule-based) to get a clean subset quickly. If you see obvious cloud contamination in visual checks or validation metrics, add **Option B** or **Option C**.

Combined Binary + Severity Training Workflow

Here we describe the two-phase training workflow for wildfire detection and severity assessment:

1. **Phase 1**: Train a binary fire model on combined CEMS + Sen2Fire data (with vegetation/NDVI)
2. **Phase 2**: Fine-tune the severity head on CEMS GRA data only (with vegetation/NDVI)

Phase	Data	Model	Output
1	CEMS DEL + Sen2Fire (combined)	Single-head binary (2 classes)	Binary fire detection model
2	CEMS GRA only	Dual-head (binary frozen + severity trained)	Binary + severity model

This approach maximizes geographic diversity for fire detection (Europe + Australia) before specializing on severity (CEMS-only, since Sen2Fire has no severity labels). **Input channels**: 8 (7 spectral bands + NDVI) for both phases.

Why this workflow

Aspect	Benefit
Phase 1 – Combined binary	More fire examples from two continents (Europe + Australia); better generalization
Phase 2 – Severity on CEMS	Severity (GRA) exists only in CEMS; fine-tune a dedicated head without diluting binary performance
Two-phase separation	Binary head learns from all data; severity head learns only where labels exist
Vegetation (NDVI)	Helps separate burn scars from water, shadow, and soil; improves both detection and severity

Vegetation (NDVI)

NDVI (Normalized Difference Vegetation Index) is used as the 8th input channel:

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$$

- **High NDVI** → healthy vegetation
- **Low NDVI** → burned areas, water, bare soil, shadow

Both CEMS and Sen2Fire support 8-channel input (7 bands + NDVI). Use NDVI by default; disable with *–no-ndvi* if needed.

Phase 1: Combined Binary Training

Goal: Train a binary fire detection model on CEMS + Sen2Fire.

Data

- **CEMS**: DEL patches (binary mask: 0=no fire, 1=fire)
- **Sen2Fire**: Binary patches (scene1 + scene2 for train; scene3 for val; scene4 for test)
- **Combined**: ConcatDataset of both loaders

Model

- Single-head binary (2 classes)
- 8 input channels (7 bands + NDVI)
- Same architecture as the binary head of FireDualHeadModel (encoder + decoder + segmentation head)

```
# 1. Generate CEMS DEL patches (if not already done)
uv run python run_pipeline.py \
    --dataset-dir ../wildfires-cems \
    --output-dir ./patches \
    --mask-type DEL

# 2. Ensure Sen2Fire data is available (scene1, scene2, scene3, scene4)

# 3. Train combined binary model (script: train_combined_binary.py)
uv run python train_combined_binary.py \
    --patches-dir ./patches \
    --sen2fire-dir ../data-sen2fire \
    --output-dir ./output/combined_binary \
    --epochs 50
```

Output: Checkpoint with encoder, decoder, and binary head. No severity head yet.

Phase 2: Severity Fine-Tuning

Goal: Add a severity head and train it on CEMS GRA data.

- **Data**
  - **CEMS GRA patches only** (5 severity classes: no damage, negligible, moderate, high, destroyed)

    - Generate with: run_pipeline.py --mask-type GRA --output-dir ./patches_gra
- **Model**
    - FireDualHeadModel: binary head (from Phase 1) + severity head (new)
    - **Frozen**: encoder, decoder, binary head
    - **Trained**: severity head only

Training

```
# 1. Generate CEMS GRA patches (if not already done)
uv run python run_pipeline.py \
    --dataset-dir ../wildfires-cems \
    --output-dir ./patches_gra \
    --mask-type GRA

# 2. Fine-tune severity head (script: train_severity_finetune.py)
uv run python train_severity_finetune.py \
    --checkpoint ./output/combined_binary/checkpoints/best_model.pt \
    --patches-dir ./patches_gra \
    --output-dir ./output/severity_finetune \
    --epochs 30
```

Output: Dual-head checkpoint (binary + severity). Use with inference and the app for both fire detection and severity maps.

Two-Head Model Rationale

Phase 2 uses a **2-head model** (FireDualHeadModel) because:

1. **Inference**: One model provides both binary fire detection and severity maps.
2. **Hierarchy**: Severity is only meaningful where fire exists; binary acts as a gate.
3. **Stability**: Phase 1 binary head stays frozen; only the severity head is trained.
4. **Reuse**: Existing FireDualHeadModel and inference pipeline support this setup.

The severity head is a single Conv2d layer on top of the shared decoder output. It is randomly initialized in Phase 2 and trained only on CEMS GRA data.

Data Requirements

Dataset	Phase	Mask type	Patches
CEMS	1	DEL (binary)	./patches (train/val/test)
Sen2Fire	1	Binary	../data-sen2fire (scene1–4)
CEMS	2	GRA (severity)	./patches_gra (train/val/test)

V3 Pipeline: Model Architectures & Training Commands

Here we describe the model architectures used in the combined binary + severity workflow, recommended encoder/decoder combinations, and **copy-paste commands** to train each variant.

Patch Generation

We generate patches before training

CEMS DEL patches (../patches)

Binary fire/no-fire masks. Used for **Phase 1** (combined binary training).

```
cd fire-pipeline

uv run python run_pipeline.py \
    --dataset-dir ../wildfires-cems \
    --output-dir ../patches \
    --mask-type DEL
```

Output: ../patches/train/, ../patches/val/, ../patches/test/ with *_image.npy and *_mask.npy (binary 0/1)

CEMS GRA patches (../patches_gra)

What is patches_gra? CEMS patches with **GRA (Grading)** masks — 5 severity levels (no damage, negligible, moderate, high, destroyed) instead of binary. Used for **Phase 2** (severity fine-tuning). Only CEMS images that have GRA annotations are included (check satelliteData.csv column GRA).

```
cd fire-pipeline

uv run python run_pipeline.py \
    --dataset-dir ../wildfires-cems \
    --output-dir ../patches_gra \
    --mask-type GRA
```

Output: ../patches_gra/train/, ../patches_gra/val/, ../patches_gra/test/ with *_image.npy and *_mask.npy (values 0–4).

Sen2Fire patches (../data-sen2fire)

Sen2Fire patches are **not** generated by run_pipeline.py. The dataset is downloaded from Zenodo 10881058 so you have

```
./data-sen2fire/
├── scene1/   # .npz files (train)
├── scene2/   # .npz files (train)
├── scene3/   # .npz files (val)
└── scene4/   # .npz files (test)
```

Architecture overview

The pipeline uses **segmentation_models_pytorch (smp)** with two components:

Component	Role	Example
Encoder	Backbone for feature extraction (pretrained on ImageNet)	ResNet34, EfficientNet-B0
Decoder	Reconstructs segmentation map from encoder features	U-Net, U-Net++, DeepLabV3+

Input (8 ch) → Encoder → Decoder → Heads (binary 2-class, severity 5-class)

Summary

- **Phase 1**: Single-head binary (encoder + decoder + binary head)
- **Phase 2**: Dual-head (encoder + decoder + binary head frozen + severity head trained)

Available options for the 2-phases architecture

Encoders (backbones)

Encoder	Params (approx)	Use case
resnet18	~11M	Fast, lightweight
resnet34	~21M	Default, balanced
resnet50	~25M	Higher accuracy
efficientnet-b0	~5M	Efficient, good quality
efficientnet-b1	~7M	Stronger EfficientNet
efficientnet-b2	~9M	Best quality (EfficientNet family)
mobilenet_v2	~3.5M	Edge deployment

Decoder architectures

Architecture	Description	Best for
unet	Classic U-Net with skip connections	General purpose, robust baseline
unetplusplus	Nested skip connections, denser decoder	Fine boundaries, small burn scars
deeplabv3plus	Atrous convolutions, multi-scale context	Large burned areas, global context

Recommended Model combinations

Default / Balanced

Encoder	Architecture	Why
resnet34	**unet**	Solid baseline, moderate size, widely used. Good starting point.

2. Higher Accuracy

Encoder	Architecture	Why
resnet50	**unetplusplus**	Deeper encoder + denser skip connections. Better for fine-grained severity.
efficientnet-b1	**unetplusplus**	EfficientNet scales well; U-Net++ improves boundary detail.

3. Fast / Lightweight

Encoder	Architecture	Why
resnet18	**unet**	Smaller, faster. Good for quick experiments or limited compute.

mobilenet_v2	**unet**	Very lightweight. Suitable for edge or real-time deployment.

4. Best Quality (more compute)

Encoder	Architecture	Why
efficientnet-b2	**unetplusplus**	Strong encoder + decoder. Best for severity and small burn scars.
resnet50	**deeplabv3plus**	Atrous convolutions for multi-scale context. Good for large fires.

THE INFERENCE PROCESS

What about inference?

Inference is about using our trained model on new images. The input format for the inference will be identical to training input:

- Same 7 bands (or 12 if you use all)
- Same 256×256 patch size
- Same value range (0-1)

Here you can see an schema of the Training- Inference process

```
Training:                                Inference:

┌──────────────┐                         ┌──────────────┐
│ Image Patch  │                         │ NEW Image    │
│ (7, 256,256) │                         │ (7, 256,256) │
│      +       │                         │              │
│ Ground Truth │      ──────►            │ No labels!   │
│ Mask         │    (trained             │ Model        │
└──────────────┘     model)              │ predicts     │
                                         └──────────────┘
```

Inference on a full image

When dealing with real images, since they are huge, we will:

- Cut the new image into patches (no overlap needed, or small overlap)
- Run each patch through the model
- Stitch the predictions back together

What are the labels?

The labels are the mask files (DEL and GRA) so, for every satellite image, there is a corresponding mask that shows where fire burned. So for each satellite image

- EMSR230_AOI01_01_S2L2A.tif ← INPUT (what satellite sees)
- EMSR230_AOI01_01_DEL.tif ← LABEL (binary: fire/no fire)
- EMSR230_AOI01_01_GRA.tif ← LABEL (severity: 0-4)
- EMSR230_AOI01_01_CM.tif ← Cloud mask (helper data)

After patching:

- patch_r0_c0_image.npy ← INPUT (256×256×7 channels)
- patch_r0_c0_mask.npy ← LABEL (256×256 with 0/1 or 0-4)

How the data was labeled?

The data was not labeled by an algorithm, but by human experts at Copernicus Emergency Management Service (CEMS), as follows:

1. Fire event reported (e.g., Portugal 2017)
2. Analysts get PRE-fire satellite image
3. Analysts get POST-fire satellite image
4. Analysts MANUALLY draw polygons around burned areas (comparing pre vs post, using spectral signatures)
5. Polygons converted to raster masks (DEL, GRA)

That's the reason why the labels are high quality — they're human-verified ground truth, not automated predictions.

What the model learns

The model learns the relationship between spectral patterns and fire presence: when pixel looks THIS, then label THAT.

The training data (from satellite images) will include masks (labels as DEL/GRA), it's acquired after the fire (post-fire), where the burned area is visible and its quality is based on curated data (cloud free selected)

Then, with the inference (here we try to use as close as possible images from real time datasets) we provide a prediction of fire detection and severity. The data for inference will be acquired during fire or right after fire.

Which challenges we could face with Real Time data?

When we use our trained model on new satellite images, we could face the following challenges:

- No mask — then one of model's jobs is to predict that mask
- If there is no curated cloud mask, then we may need to:
    - Use Sentinel-2's built-in cloud detection
    - Train a separate cloud detection model
    - Use external cloud masking service
- There is no guaranteed quality, so the images may have:
    - Partial cloud cover
    - Smoke from active fires
    - Haze or atmospheric interference
- There is no geographical metadata… so we would need to track coordinates.

Schema of the inference pipeline to be built.

```
Real-time Sentinel-2 image
        ↓
┌──────────────────────┐
│ 1. Download from ESA │   (Copernicus Open Access Hub)
│    Sentinel-2 API    │
└──────────────────────┘
        ↓
┌──────────────────────┐
│ 2. Select same 7 bands │   indices (1,2,3,7,8,10,11)
│    Normalize to 0-1    │
└──────────────────────┘
        ↓
┌──────────────────────┐
│ 3. Generate cloud mask │   (optional but recommended)
│    or estimate clouds  │
└──────────────────────┘
        ↓
┌──────────────────────┐
│ 4. Cut into 256×256  │   stride=256 (no overlap needed)
│    patches           │
└──────────────────────┘
        ↓
┌──────────────────────┐
│ 5. Run YOUR MODEL    │   → predicted mask per patch
└──────────────────────┘
        ↓
┌──────────────────────┐
│ 6. Stitch patches back │   → full image prediction
│    into full image     │
└──────────────────────┘
        ↓
    Fire/burn map!
```

Metrics

CombinedMetrics

- **SegmentationMetrics**: Confusion matrix over pixels; then per-class and aggregate:
    - **IoU**, **Dice**, **Precision**, **Recall** per class.

    - **mean_iou**, **mean_dice**.
    - **fire_iou**, **fire_dice**, **fire_precision**, **fire_recall** (all classes with index > 0 pooled).
- **DetectionMetrics**: Patch-level. "Has fire" = any pixel predicted as fire; same for target. Then **accuracy**, **precision**, **recall**, **F1** (and tp/fp/tn/fn).

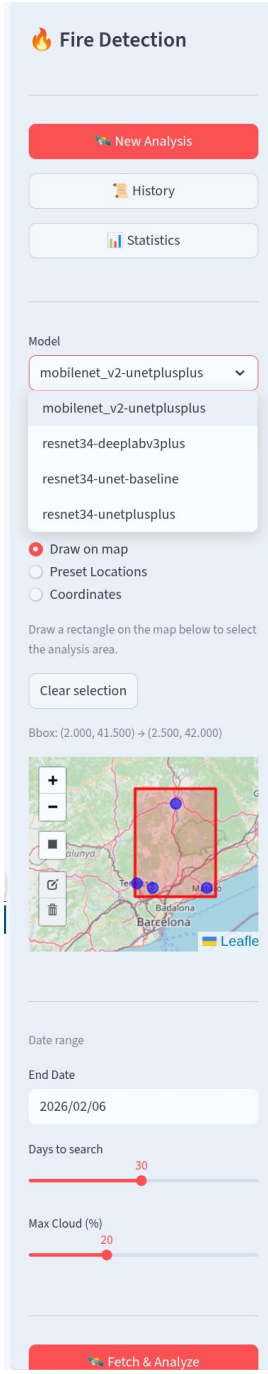
What is used for model selection and logging

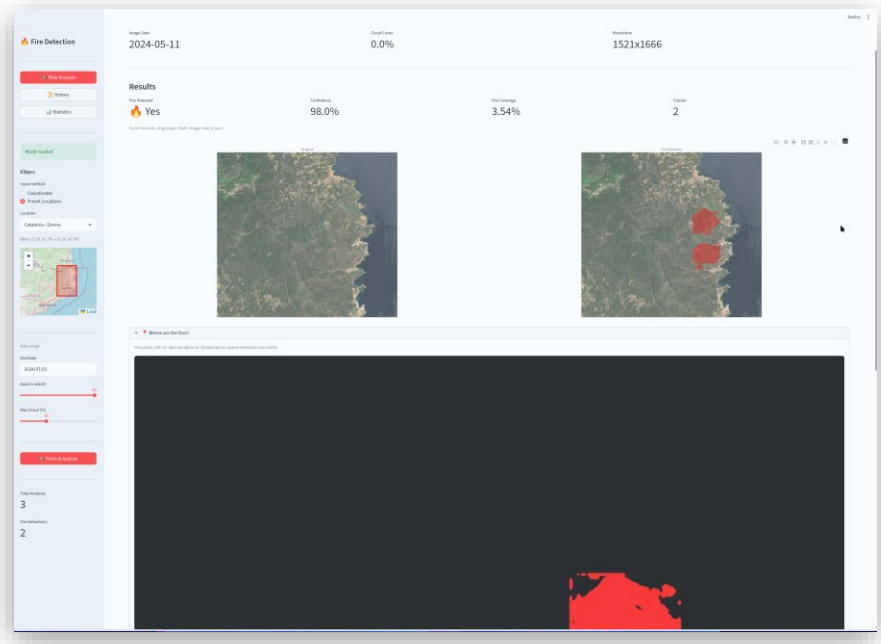
- **Best checkpoint**: Saved when **validation fire_iou** improves. This is the **primary** metric.
- **Scheduler**: Steps on **val fire_iou** (maximize).
- **Printed each epoch**: Train/val loss; val fire_iou, fire_recall, detection_f1.
- **W&B** (if enabled): train/val loss, train/val fire_iou, train/val detection_f1, val fire_recall, learning rate.

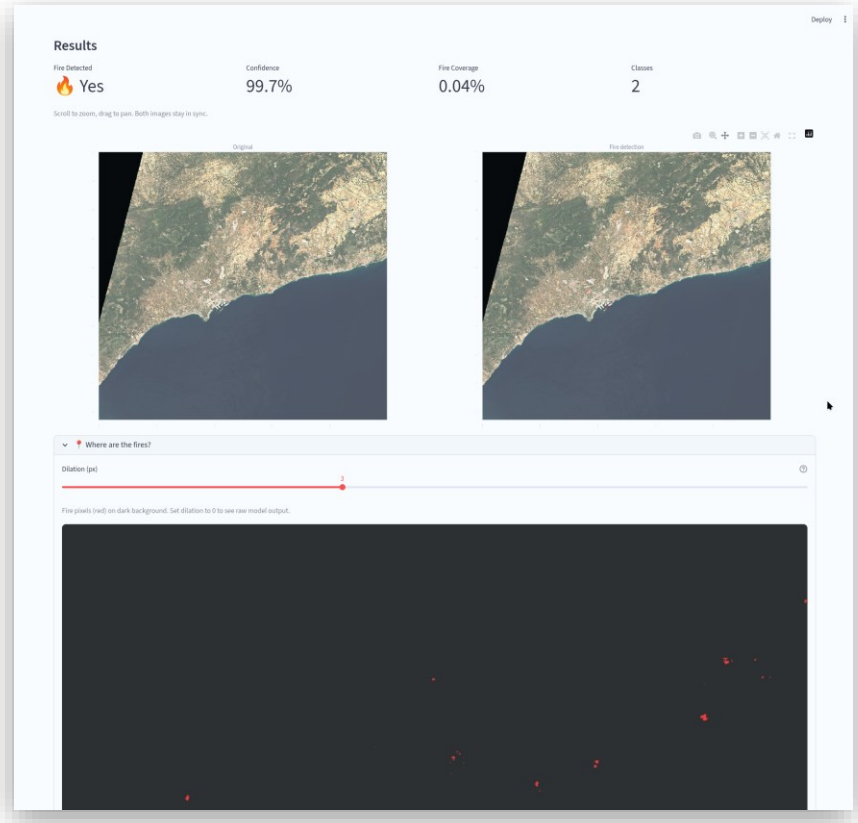
Talking point: "We optimize for fire IoU; detection F1 and recall are monitored but the saved best model is by fire IoU."

Application design and features.

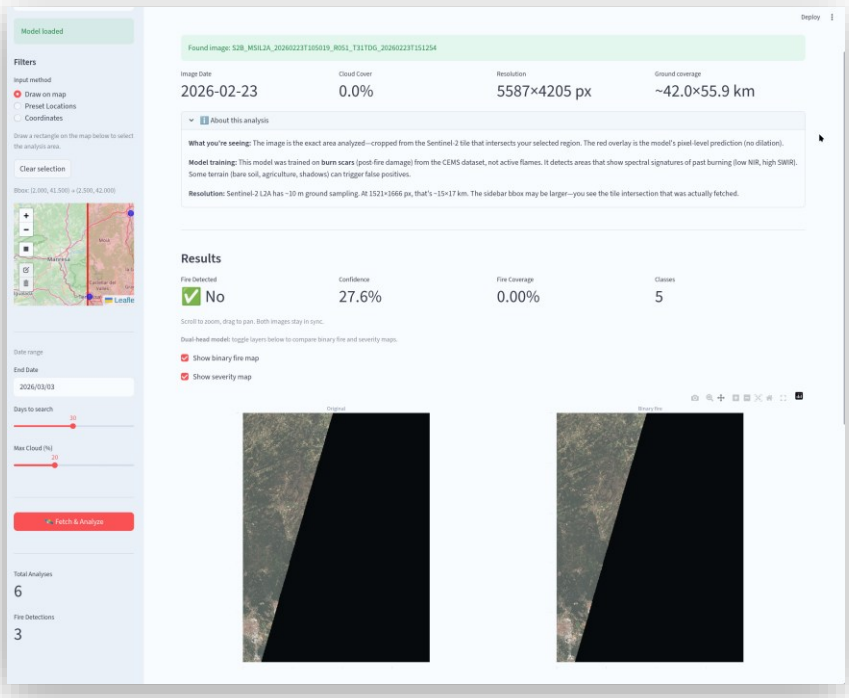
The application is intended to display both severity maps (based on the 1st dataset training) and binary maps (based on the 2nd dataset fine tuning) as layers - so one could visually compare for ex the maps.



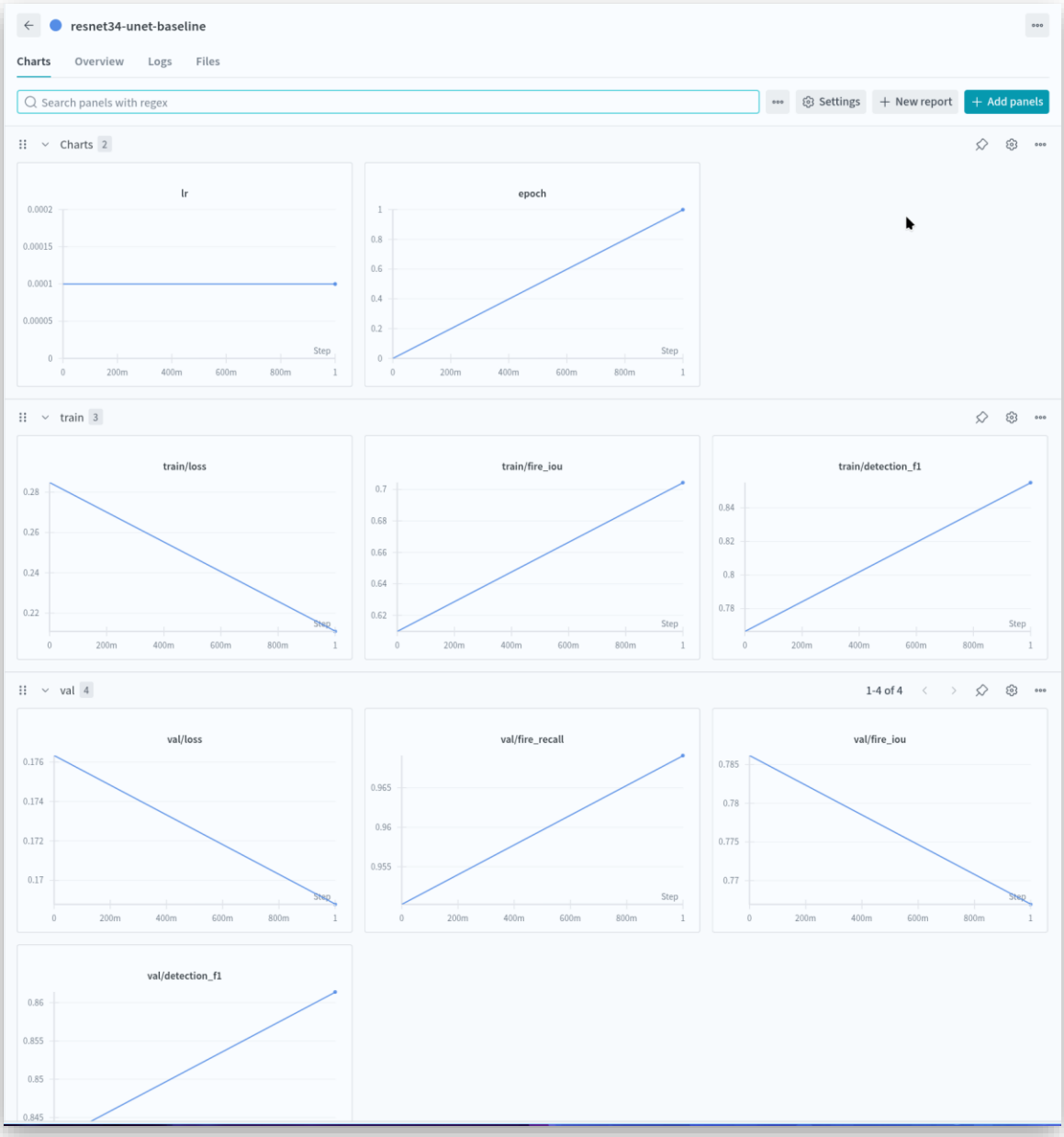

Fire Detection
2024-05-11
0.0%
1521x1666
Results
Yes
98.0%
3.54%
2
Where are the fires?


Deploy
Results
Fire Detected
Yes
Confidence
99.7%
Fire Coverage
0.04%
Classes
2
Scroll to zoom, drag to pan. Both images stay in sync.
Original
Fire detection
Where are the fires?
Dilation (px)
Fire pixels (red) on dark background. Set dilation to 0 to see raw model output.

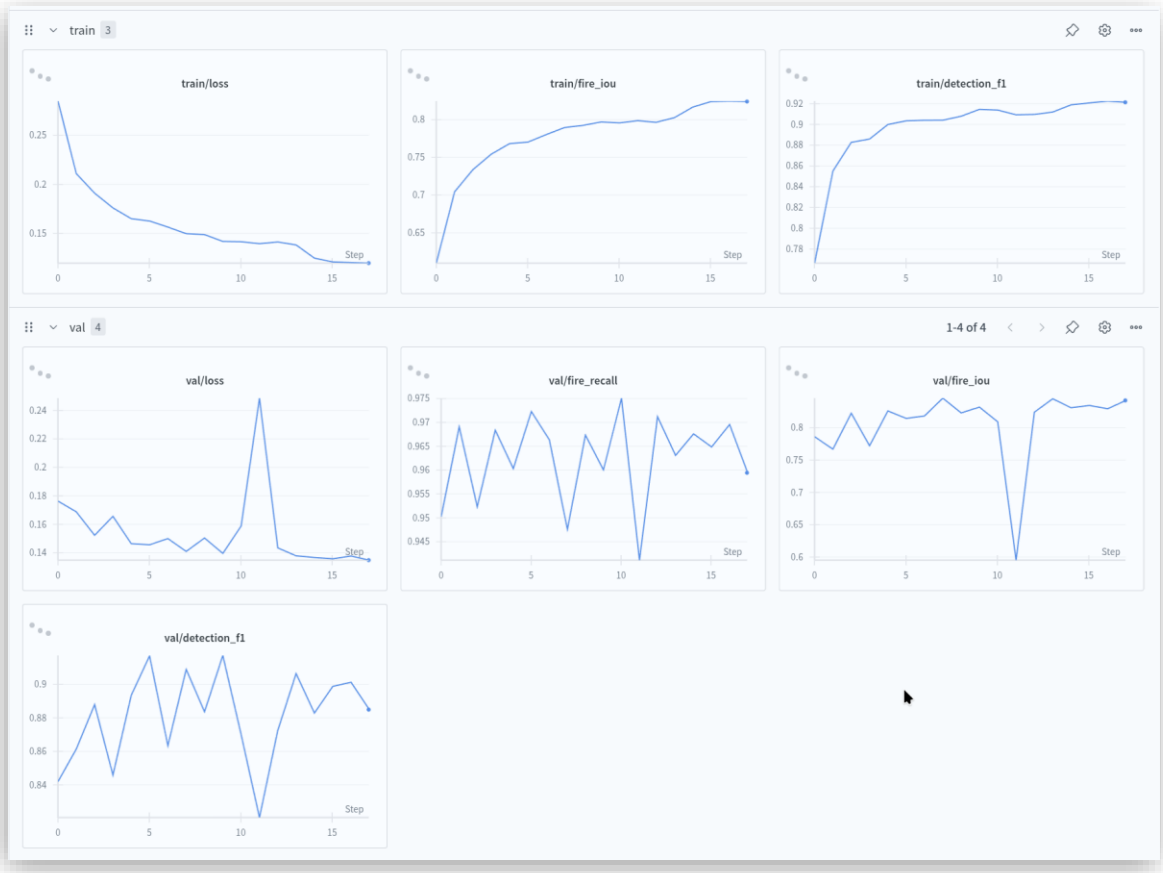
In this case, the application shows no fires (usual in winter's time)

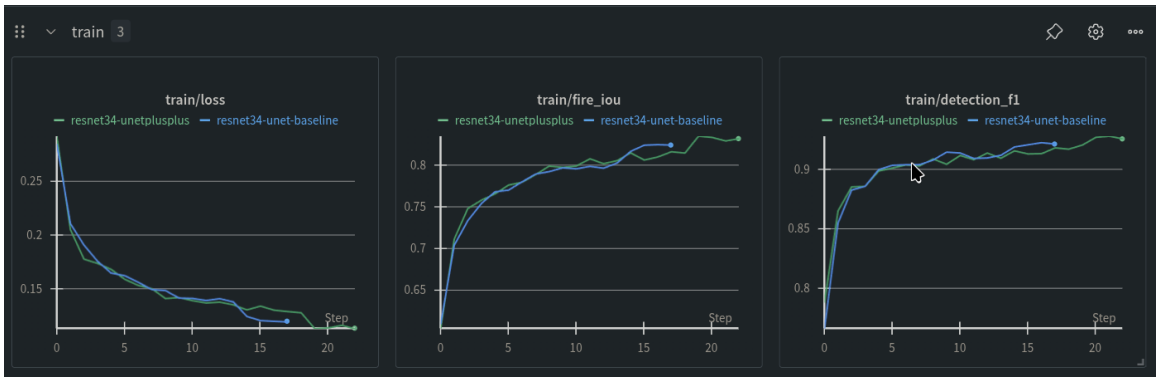


Machine Learning Ops: Testing and Performance / Weights and Biases analysis



train 3
train/loss
train/fire_iou
train/detection_f1
Step


train 3
train/loss
train/fire_iou
train/detection_f1
val 4
1-4 of 4
val/loss
val/fire_recall
val/fire_iou
val/detection_f1
Step


train 3
train/loss
resnet34-unetplusplus
resnet34-unet-baseline
0.25
0.2
0.15
Step
0
5
10
15
20
train/fire_iou
resnet34-unetplusplus
resnet34-unet-baseline
0.8
0.75
0.7
0.65
Step
0
5
10
15
20
train/detection_f1
resnet34-unetplusplus
resnet34-unet-baseline
0.9
0.85
0.8
Step
0
5
10
15
20

Analysis of results, discussion and next steps.

Results: brief description/explanation of the results.

Conclusions

Conclusions: what insights do you get from the results (usually these lead to new hypothesis)

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